Reimagining the Galaxy Computational Workbench in the Age of AI

Jeremy Goecks
Moffitt Cancer Center & University of South Florida
November 2026





Community

Training,
Publications,
Communication &
Collaborations

Interfaces

Web GUI
APIs & CLI
Agents and MCP

Galaxy PROJECT

Data

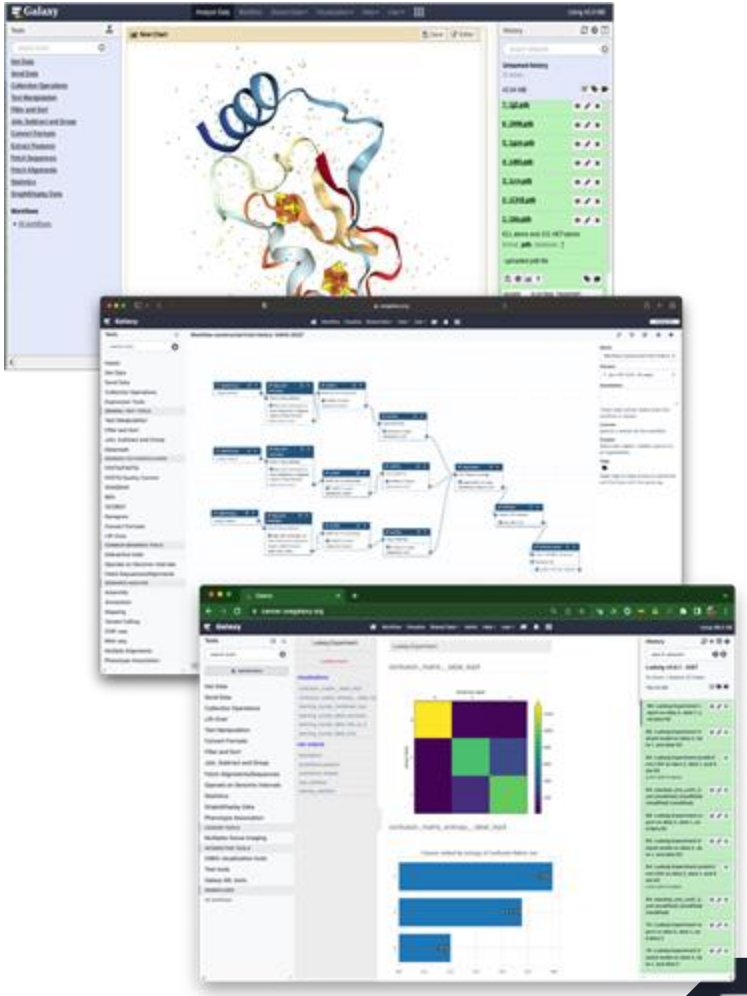
Reference
Data
Data Libraries
User Data

Tools & Workflows

Interactive tools
Visualizations
bioconda &
biocontainers

Compute

usegalaxy.*
On-prem
Laptops
Cloud



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738,900+ registered users
24,700+ citations
9,400+ institutions
163+ countries/territories
13,000+ tools integrated



Interfaces

Agentic UI



Web UI

Programmatic API

Datasets

Galaxy

Computing Resources

HPC clusters

Cloud

High-throughput

Analysis Tools and Visualizations

Quality control

Image processing

Drug response prediction

Interactive charts

Network graphs

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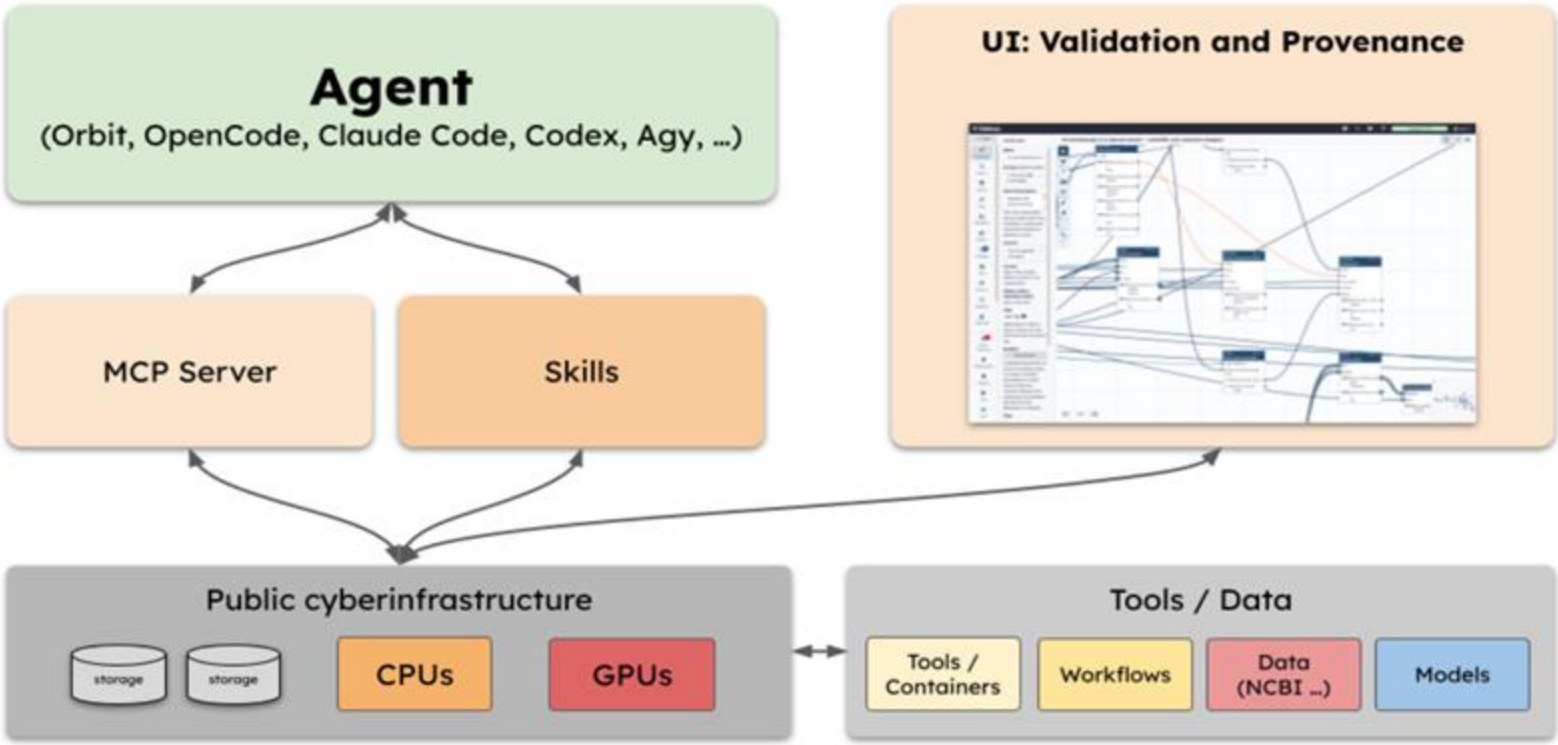
Rearchitecting Galaxy for Agents



Dannon Baker



Anton Nekrutenko



Baker D, ..., Nekrutenko A. Verifiable Agentic Science on Public Infrastructure. bioRxiv (Oct 2026). doi:10.64898/2026.10.26.76 1864

Galaxy Plugins for Agent Interfaces



Junhao Qiu
Software Engineer

Claude

Rerun the image learner tool with an increased number of epochs and the resnet18 architecture

Used Galaxy integration, loaded tools

Loading the Image Learner tool's parameters.

Searched available tools run_tool get_tool_input_template search_tools_by_name

Get job details

Checking the latest run's architecture against the request.

Get tool input template

Deciding whether to rerun the job.

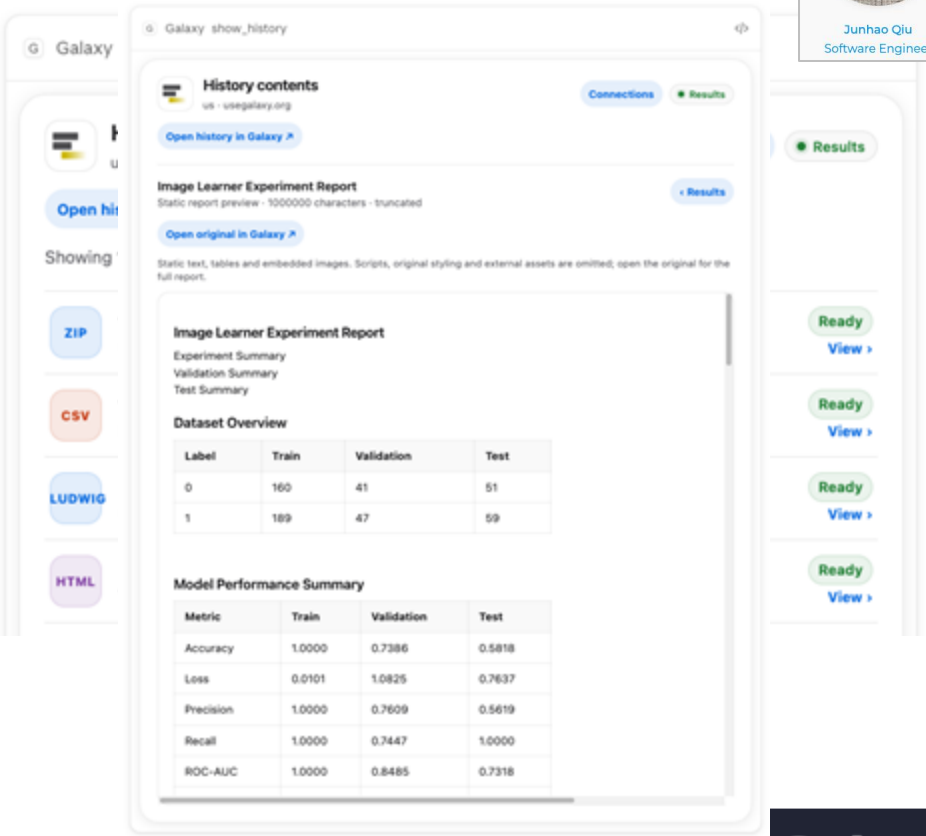
Run tool

I submitted the Image Learner job in the same history. It's accepted but not finished (request ID `1fac9e00ce6af212`).

The most recent earlier run in that history already used resnet18 for 30 epochs. So the only real change is the epoch count, which I raised from 30 to 60. Say so if you want a different number.

Everything else matches that earlier run:

- **Model:** resnet18, pretrained and fine-tuned, with the image size set to 224x224
- **Task:** binary, validated on ROC AUC, with the threshold tuned on balanced accuracy
- **Training:** early stop at 5, random seed 2026, 70/10/20 train/validation/test split
- **Inputs:** `fullgrid_rgb_16384_images.zip` and `fullgrid_rgb_16384_metadata.csv`



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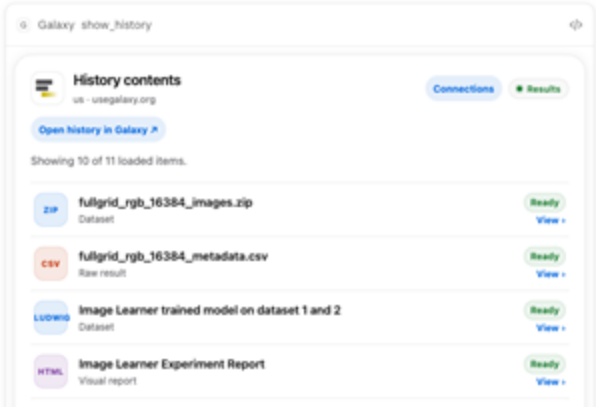
Run tool

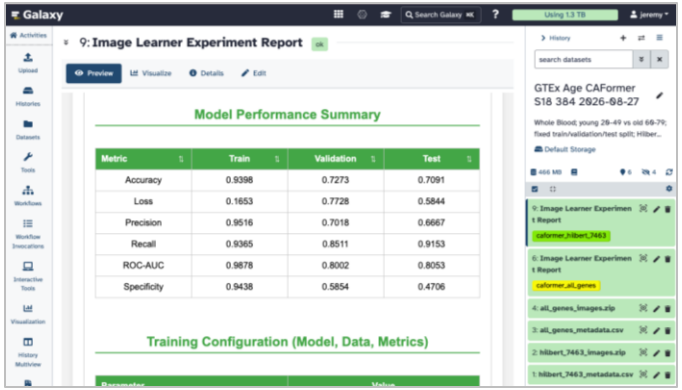
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Junhao Qiu
Software Engineer

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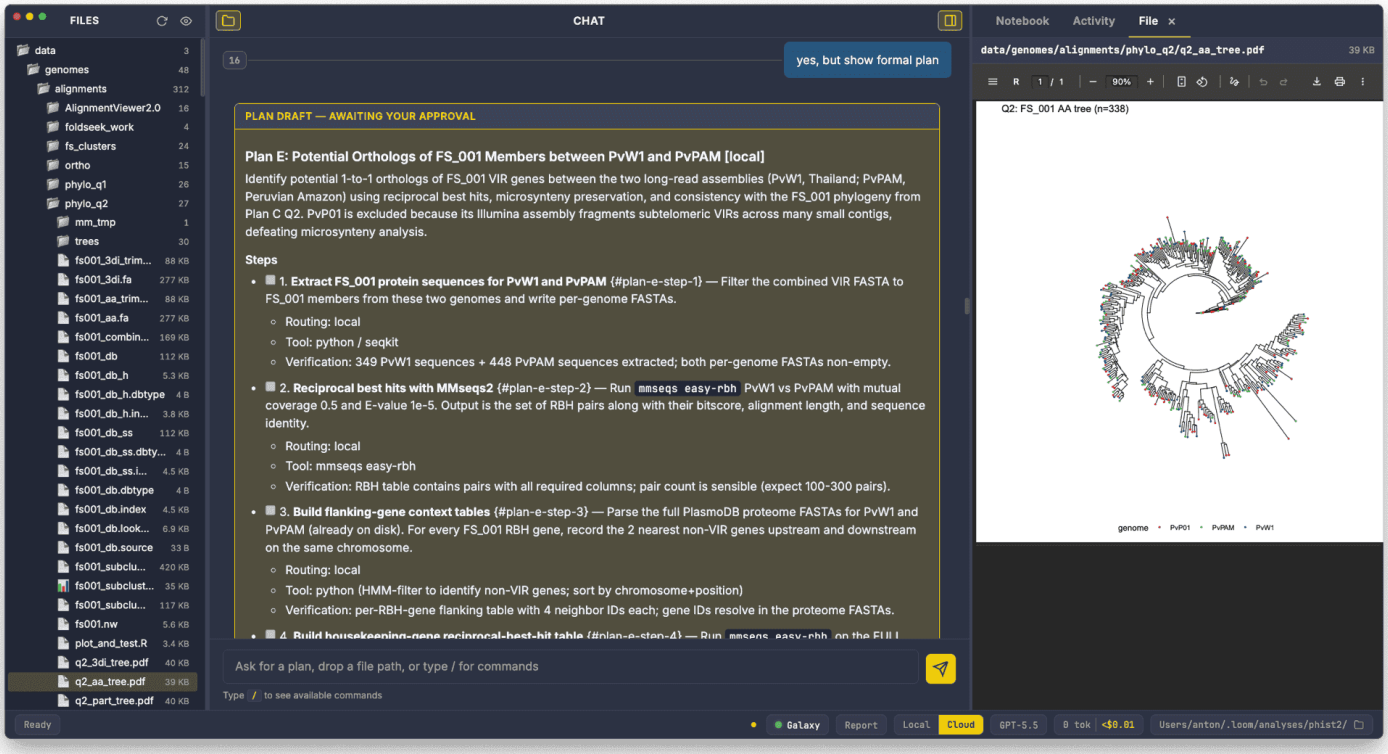
Orbit: an Agentic AI App for Galaxy

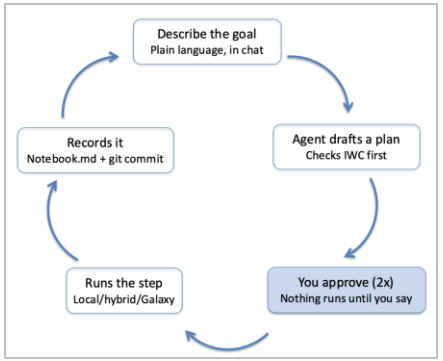


Dannon Baker



Anton Nekrutenko





Poster 126: Orbit—Reproducible agentic workflow development with cloud execution included

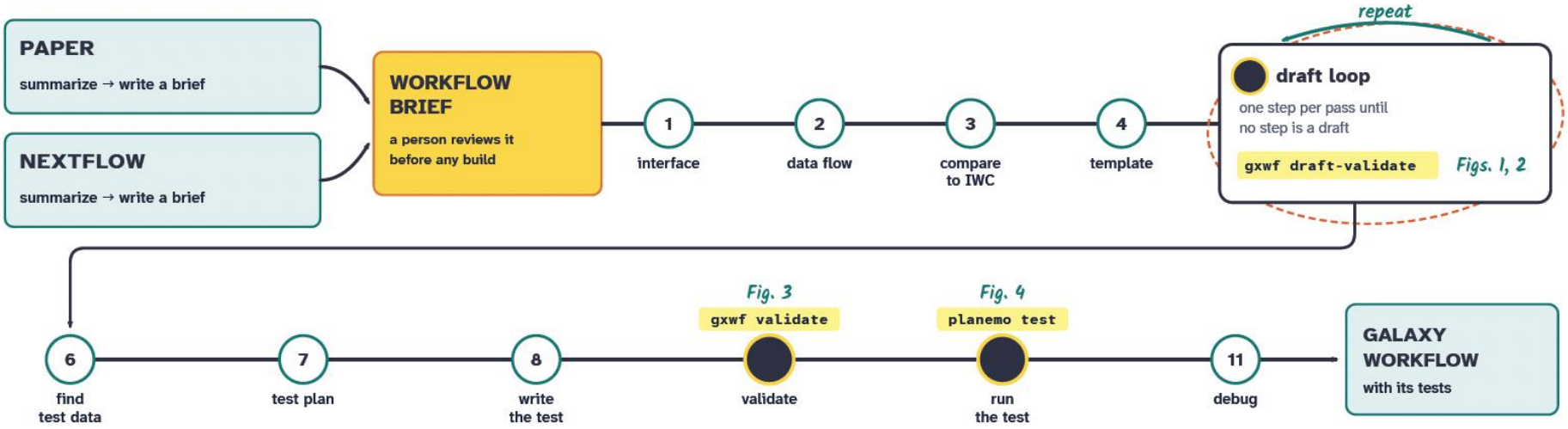
Anton Nekrutenko, Scott Cain, Dannon Baker.

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Foundry: Helping Agents Turn Papers and Pipelines into Galaxy Workflows



John Chilton



https://galaxyproject.github.io/foundry/

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Benchmarking Agent Use of Galaxy





4
Models

- GPT-5.5
- GPT-5.6 Sol
- GPT-5.6 Luna
- DeepSeek V4 Pro

×

2
Execution conditions

- **Galaxy** (workflow-based)
- **Custom code**

×

3
Benchmarks

- **BixBench-Verified-50** (50 tasks)
- **CompBioBench** (100 tasks)
- **IWC-WorkflowBench** (10 tasks)

3 replicates per task

3,840 runs

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Three Diverse Bioinformatics Benchmarks

BixBench-Verified-50[1]
50 tasks

Expert-verified subset of BixBench: open-answer data-analysis questions drawn from published analysis notebooks across 33 datasets

Examples

- Find the gene whose expression is most negatively correlated with CRISPR essentiality across cell lines
- Answer phylogenetics questions that require PhyKIT or BUSCO HMM databases

CompBioBench[2]
100 tasks

Diverse genomics, epigenomics, transcriptomics, machine learning, and single-cell tasks

Examples

- Identify the species whose RNA-seq reads were mixed into a human sample
- Run OpenSpliceAI on a portion of the human genome to predict donor and acceptor splice sites plus overall gene structure

IWC-WorkflowBench[3]
10 tasks

Agents reproduce analyses from peer-reviewed, tested Galaxy workflows curated by the Galaxy community

Examples

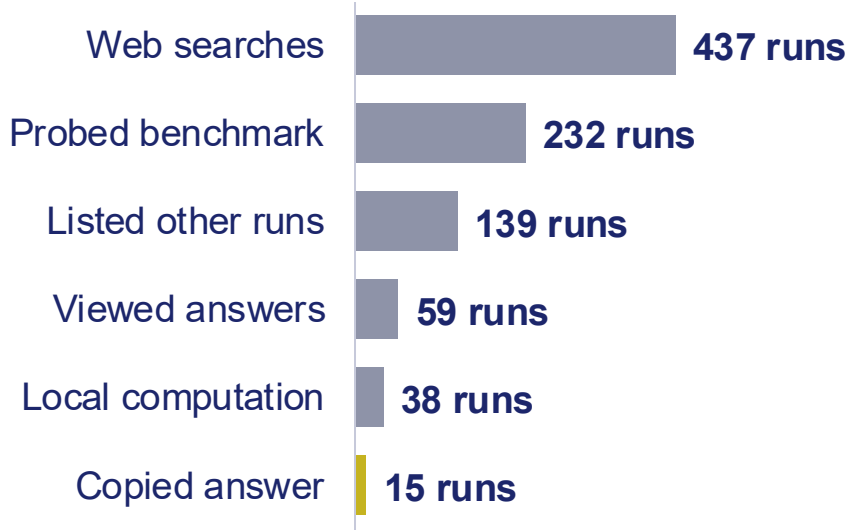
- Assemble genome from HiFi reads, estimate genome size and heterozygosity from k-mers
- Process multiplex tissue imaging dataset to segment nuclei, assign cell phenotypes, and characterize cellular neighborhoods

1. Mitchener et al., BixBench, arXiv:2503.00096 (2025); Phylo, BixBench-Verified-50, Hugging Face (Feb 2026)
2. Nair et al., CompBioBench, bioRxiv doi:10.64898/2026.04.06.716850 (Apr 2026)
3. Galaxy Workflow Repository: https://iwc.galaxyproject.org

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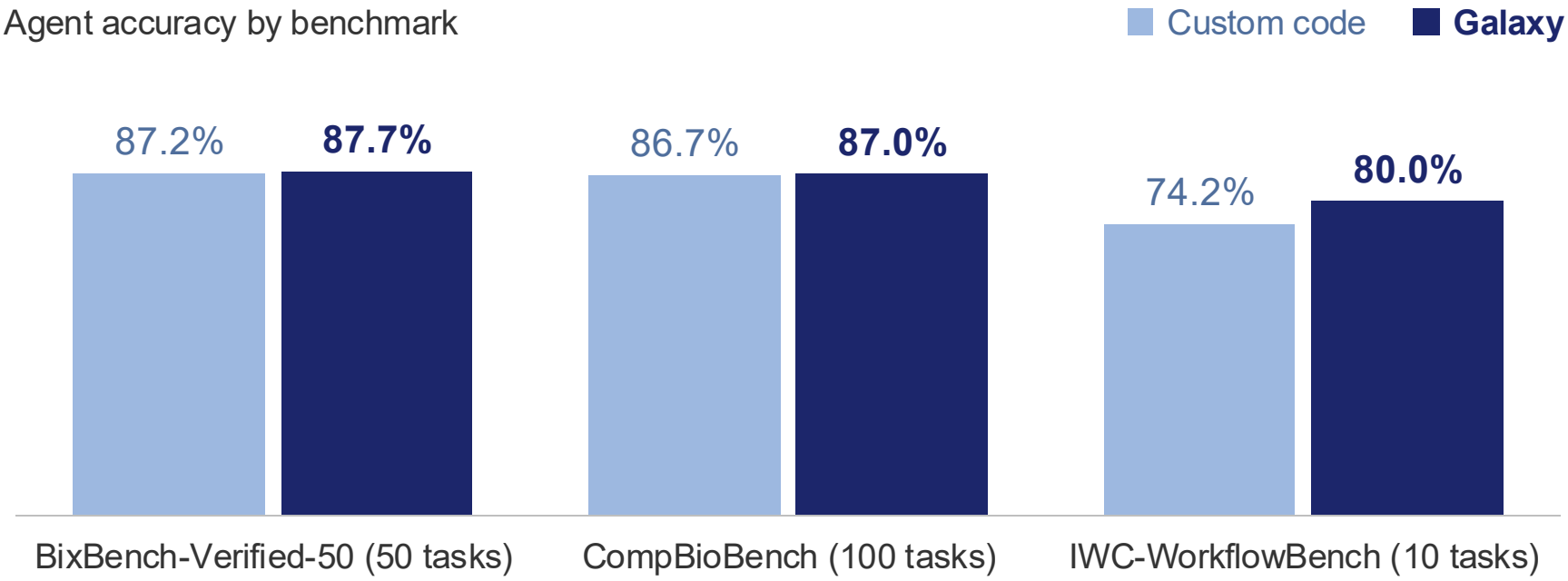
Running a Benchmark Requires Careful Monitoring of Agent Behavior

- ~25% of the time agents broke a rule
- DeepSeek accounts for 74 of the 108 major violations
- We used agents for flagging violations



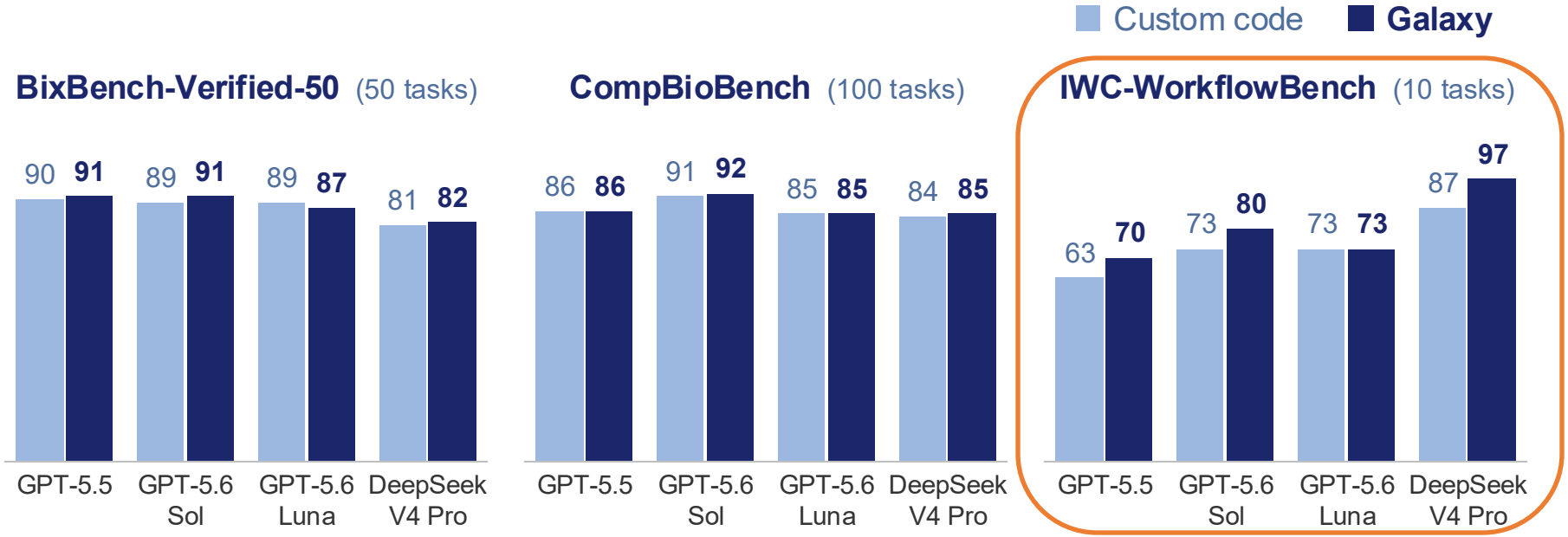
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Agent Accuracy with Galaxy Matches or Exceeds Custom Code



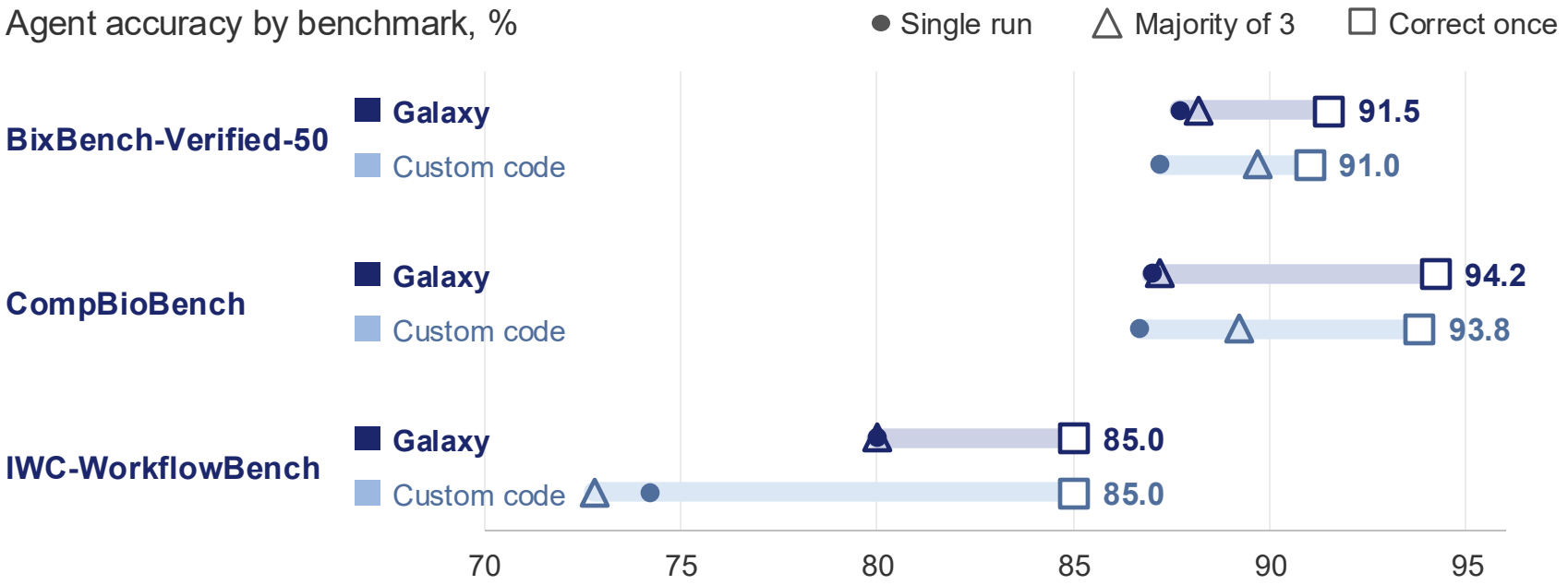
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Agent Accuracy With Galaxy Increases When Analyses are More Complex



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Consensus and Validation can Improve Agent Accuracy



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An Example of How Agents Make Mistakes

bix-52-q7 (BixBench-Verified-50). Rows removed by a >90% / <10% methylation filter. Correct: ✓ **19,159** · wrong in 3 of 12 Galaxy runs

GPT-5.6 Sol r3	✗ **19,160**	line count includes the header
GPT-5.6 Luna r2	✗ **19,160**	line count includes the header
GPT-5.6 Luna r3	✗ **539**	reported the rows kept (539), not removed

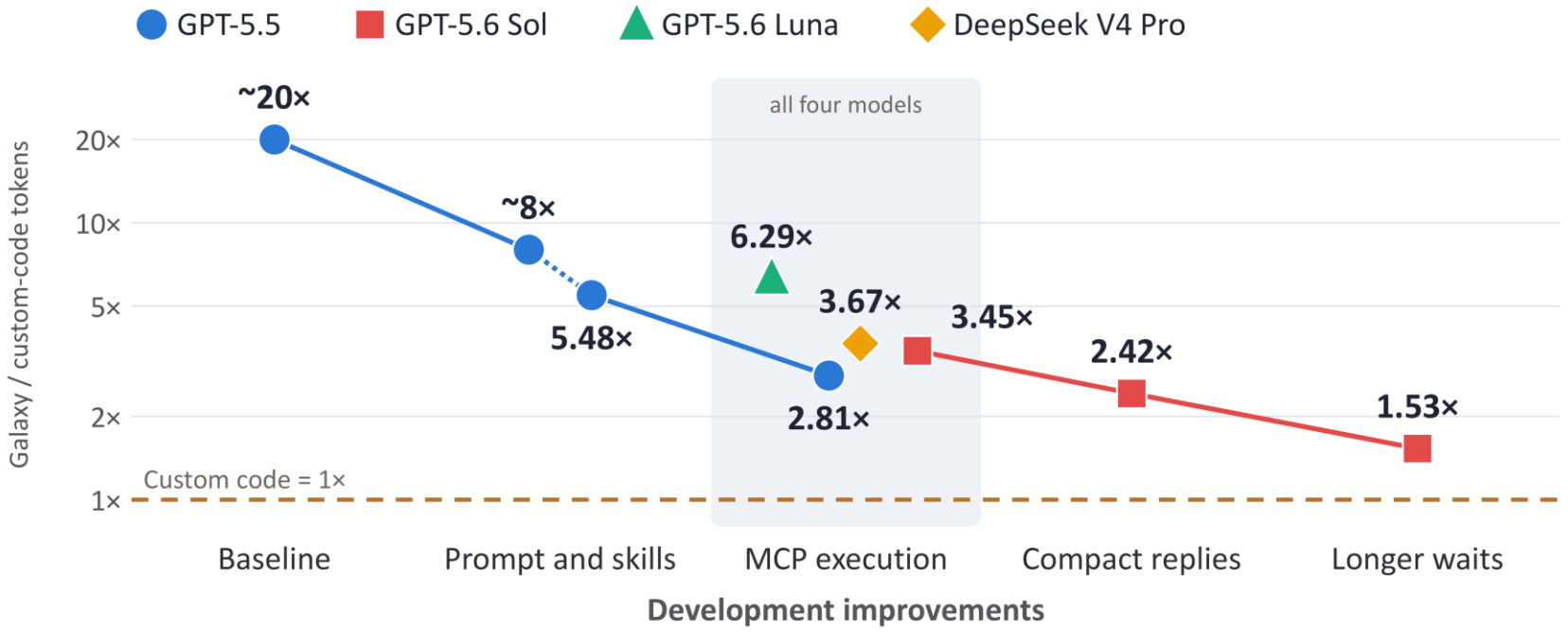
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Another Example of How Agents Make Mistakes

RUN	1 ILLUMINATION	2 STITCH + REGISTER	3 FIND TMA CORES	4 SEGMENT NUCLEI	5 QUANTIFY	6 TO ANNDATA	7 PHENOTYPE	8 SPATIAL STATS	SCORE	TIME
GPT-5.6 Sol										
Replicate 1	Illumination ✓	ASHLAR ✓	UNetCoreograph ✓	Mesmer ✓	MCQUANT ✓	MCMICRO→A ✓	Phenotyping ✓	Spatial ✓ ✓ ✓	9/9	10.7 h
Replicate 2	Illumination ✓	ASHLAR ✓	UNetCoreograph ✓	Mesmer ✓	MCQUANT ✓	MCMICRO→A ✓	Phenotyping ✓	Spatial ✓ ✓ ✓	9/9	10.0 h
Replicate 3	Illumination ✓	ASHLAR ✓	UNetCoreograph ✓	Mesmer ✓	MCQUANT ✓	MCMICRO→A ✓	Phenotyping ✓	Spatial ✓ ✓ ✓	9/9	10.4 h
GPT-5.6 Luna										
Replicate 1 trial	Illumination ✓	ASHLAR ✓	UNetCoreograph ✓	Mesmer ✓	MCQUANT ✓	MCMICRO→A ✓	Phenotyping ✓	Spatial ✓ ✓ ✓	9/9	11.0 h
Replicate 2	Illumination ✓	ASHLAR ✓	UNetCoreograph ✓	Mesmer ✓	MCQUANT ✓	MCMICRO→A ✓	Phenotypi... × log: false	Spatial × × ×	5/9	14.2 h
Replicate 3	Illumination ✓	ASHLAR ✓	UNetCoreograph ✓	Mesmer ✓	MCQUANT ×	MCMICRO→A ×	Phenotyping ✓	Spatial × × ×	5/9	21.0 h
DeepSeek V4 Pro										
Replicate 1	Illumination ✓	ASHLAR ✓	UNetCoreograph ✓	Mesmer ✓	MCQUANT ✓	MCMICRO→A ✓	Phenotyping ✓	Spatial ✓ ✓ ✓	9/9	6.6 h
Replicate 2	Illumination ✓	ASHLAR ✓	UNetCoreograph ✓	Mesmer ✓	MCQUANT ✓	MCMICRO→A ✓	Phenotyping ✓	Spatial ✓ ✓ ✓	9/9	10.1 h
Replicate 3	Illumination ✓	ASHLAR ✓	UNetCoreograph ✓	Mesmer ✓	MCQUANT ✓	MCMICRO→A ✓	Phenotyping × log: false	Spatial ✓ × ×	6/9	6.0 h

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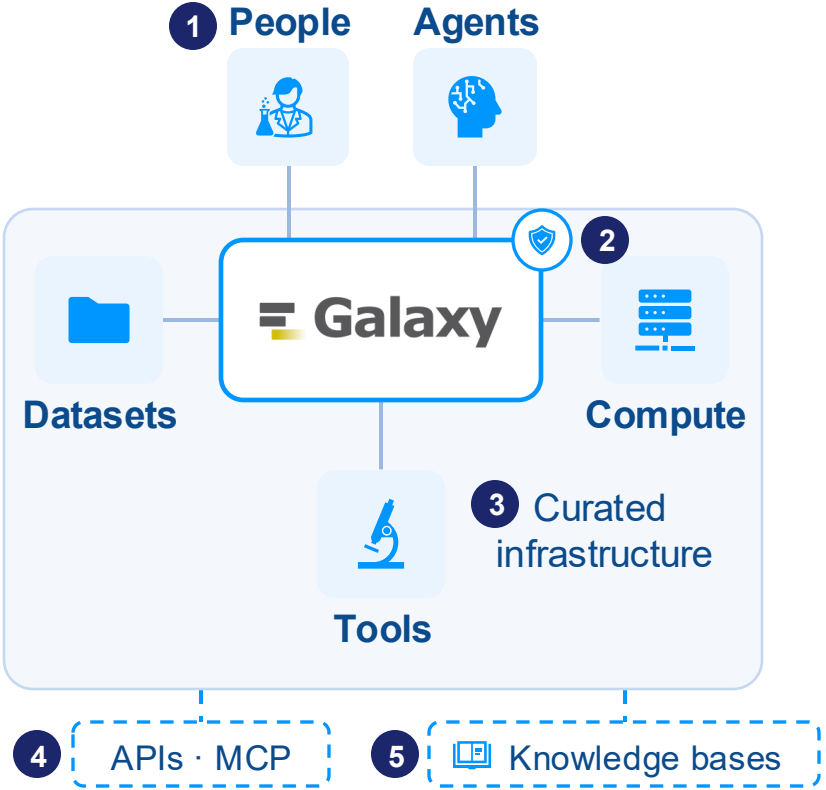
Software Engineering Reduces Agent Token Usage with Galaxy from 20x to 1.5x



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A Vision for Galaxy in an Agentic World

1. Human-centric approach through agentic collaboration
2. Promote high-quality science with reproducibility and robustness
3. Leverage curated infrastructure
4. Open and agentic-enabled
5. Integrate knowledge bases to complement analyses



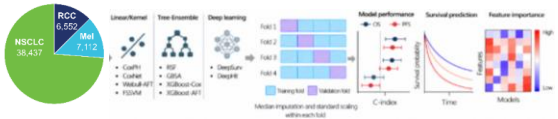
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Acknowledgements

Goecks Lab



Poster 145: Machine learning predicts cancer immunotherapy outcomes in a national real-world clinical dataset



The Galaxy Community





Funding

NIH NATIONAL CANCER INSTITUTE Informatics Technology for Cancer Research

NCI CANCER MOONSHOT Human Tumor Atlas Network A program of the National Cancer Institute of the National Institutes of Health

NSF

NIH National Human Genome Research Institute

MOFFITT CANCER CENTER

OHSU KNIGHT CANCER Institute

CHAN ZUCKERBERG INITIATIVE

SMMART Prospect Creek Foundation

THE SALAH FOUNDATION

Renaissance Philanthropy

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